

KASHYAP CHITTA

AI Research @ [ELLIS Institute Tübingen](#) and [KE:SAI](#)

✉ kashyap@kesai.eu 🌐 <https://kashyap7x.github.io> 📍 Tübingen, Germany, 72076

Employment

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| 2026 – Now | ELLIS Institute Tübingen, Germany
<i>Principal Investigator, Physical AI Lab</i> <ul style="list-style-type: none">● KE:SAI, Germany
<i>Co-Founder & Chief Technology Officer</i><ul style="list-style-type: none">• Research on visual reasoning and causal world models.• <i>Highlight:</i> Building a fully-open source compute and data-efficient self-driving stack. |
| 2025 – 2026 | NVIDIA, Germany
<i>Postdoctoral Researcher; Autonomous Vehicle Research Group</i>
University of Tübingen, Germany
<i>Visiting Researcher; Autonomous Vision Group</i> <ul style="list-style-type: none">• Research on simulation-based training and evaluation of Physical AI systems.• <i>Highlight:</i> Open-sourced the Physical AI Autonomous Vehicles dataset. |

Education

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| 2019 – 2025 | University of Tübingen, Germany
<i>PhD in Computer Science; Autonomous Vision Group</i> <ul style="list-style-type: none">• Advisor: Prof. Andreas Geiger• Thesis: Towards Scalable Autonomous Driving |
| 2017 – 2018 | Carnegie Mellon University, USA
<i>Master of Science in Computer Vision</i> <ul style="list-style-type: none">• Advisor: Prof. Martial Hebert• Thesis project: Exploiting Synthetic Data for Street Scene Segmentation• GPA: 4.15/4.33 |
| 2013 – 2017 | RV College of Engineering, India
<i>Bachelor of Engineering in Electronics and Communication</i> <ul style="list-style-type: none">• Thesis project: Monocular Visual SLAM with a Rotating Mirror• GPA: 9.11/10.0 |

Awards

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| 2026 | <ul style="list-style-type: none">• Our adaptation of Alpamayo 1.5 ranked third on the 2026 KITScenes Long Tail End-to-End Driving Challenge. |
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2025	• Our approach VaVAM-ECO ranked first on the 2025 RealADSim Closed-Loop Driving Challenge. • Our approach DiffusionLTF ranked second on the 2025 Waymo Vision-based End-to-End Driving Challenge. • Our approach SHRED ranked third on the 2025 Waymo Scenario Generation Challenge.
2024	• I was named an outstanding reviewer at ECCV 2024 (198/7293 reviewers, top 3%). • Our approach GenDM ranked second on the 2024 Dataset Distillation Challenge generative track and won the best paper award at the challenge's ECCV workshop. • Our approach TF++ ranked first on the 2024 CARLA AD Challenge map track (40 participating teams).
2023	• Our approach TF++ ranked second on the 2023 CARLA AD Challenge (20 participating teams). • I was named a top reviewer at NeurIPS 2023 (1196/11725 reviewers, top 10%). • I was named an outstanding reviewer at ICCV 2023 (130/7000 reviewers, top 2%). • I was selected for the doctoral consortium at ICCV 2023 (38 accepted participants). • Our approach PDM ranked first on the 2023 nuPlan Planning Challenge (52 participating teams). • I was named an outstanding reviewer at CVPR 2023 (232/7000 reviewers, top 3%). • I was selected as a 2023 RSS Pioneer, (30/135 applicants, 22% acceptance rate).
2022	• Our approach MapTF++ ranked first on the 2022 CARLA AD Challenge map track.
2021	• Our approach TransFuser ranked second on the 2021 CARLA AD Challenge, (100+ participating teams). • Our new computer vision lecture won the 2021 CS teaching award at the University of Tübingen.
2020	• Our approach NEAT ranked second on the 2020 CARLA AD Challenge (45 participating teams).

Internships

Jan 2019 – Aug 2019	NVIDIA, Santa Clara, USA <i>Deep Learning Intern; Autonomous Vehicles Applied Research</i> • <i>Mentor:</i> Dr. José M. Álvarez • <i>Role:</i> Research and development of an automatic dataset curation engine for the internal AI training and inference infrastructure, involving collaborations across multiple groups, which resulted in two publications.
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May 2018 –
Aug 2018

NVIDIA, Santa Clara, USA

Software Intern; [Autonomous Vehicles Applied Research](#)

- *Mentors:* [Dr. José M. Álvarez](#), [Dr. Adam Lesnikowski](#)
- *Role:* Research on approximating Bayesian Neural Networks for Active Learning which resulted in a publication, and was subsequently incorporated into the data annotation platform for the autonomous vehicles group.

Supervision

2019 – Now

University of Tübingen, Germany

Master Thesis Advisor

- *Apr 2025 – Nov 2025:* [Micha Fauth](#) (Thesis: [Evaluating Traffic and Scenario Generation using Fidelity and Diversity Metrics](#))
- *May 2025 – Oct 2025:* [Long Nguyen](#) (Thesis: [Addressing the Fundamental Barriers towards End-to-End Driving in Simulation](#))
- *Nov 2024 – Apr 2025:* [Jens Beißwenger](#) (Thesis: [Enhancing Model-Based Reinforcement Learning for Autonomous Driving](#))
- *Sep 2024 – Feb 2025:* [Melanie Schneider](#) (Thesis: [Generative Dataset Distillation: A New Hope?](#))
- *Mar 2024 – Sep 2024:* [Julian Zimmerlin](#) (Thesis: [Tackling CARLA Leaderboard 2.0 with End-to-End Imitation Learning](#))
- *Feb 2023 – Aug 2023:* [Daniel Dauner](#) (Thesis: [Vehicle Motion Planning using Data-Driven Simulation](#))
- *Dec 2022 – Jun 2023:* [Luis Winckelmann](#) (Thesis: [LiDAR-based Object Detection for Planning Transformers](#))
- *Dec 2022 – Jun 2023:* [Tim Schreier](#) (Thesis: [On Offline Evaluation of 3D Object Detection for Autonomous Driving](#))
- *Nov 2022 – May 2023:* [Siddharth Ramrakhiani](#) (Thesis: [Vision Transformers for Autonomous Driving](#))
- *Nov 2022 – May 2023:* [Jovan Cicvaric](#) (Thesis: [Generative Dataset Distillation](#))
- *Mar 2021 – Sep 2021:* [Bernhard Jaeger](#) (Thesis: [Expert Drivers for Autonomous Driving](#))
- *Oct 2020 – Apr 2021:* [Micha Schilling](#) (Thesis: [Visual Abstractions for Autonomous Driving](#))

Research Project Advisor

- *Nov 2025 – Apr 2026:* [Christopher Lindenberg](#) (Project: [Value Diffusion World Models](#))
- *Jun 2024 – Oct 2024:* [Zhengyu Su](#) (Project: [Dataset Distillation for Autonomous Driving](#))
- *Nov 2023 – Apr 2024:* [Jens Beißwenger](#) (Project: [PDM-Lite: A Rule-Based Planner for CARLA Leaderboard 2.0](#))
- *Apr 2022 – Sep 2022:* [Alexander Braun](#) and [Luis Winckelmann](#) (Project: [Infraction Visualization and Clustering for Better Agent Evaluation in CARLA](#))

Teaching

2019 – Now

University of Tübingen, Germany

Lead Teaching Assistant

- Apr 2023 – Jul 2023: [Autonomous Vision](#) (seminar, 5 teams of 2 students)
- Apr 2022 – Jul 2022: [Autonomous Vision](#) (seminar, 8 teams of 2 students)

Teaching Assistant

- Apr 2021 – Jul 2021: [Computer Vision](#) (lecture, 150 students)
- Oct 2019 – Feb 2020: [Self-Driving Cars](#) (lecture, 80 students)

Academic Activities

Workshop Organization

- CVPR 2026: [Simulation for Autonomous Driving](#), 03.06.2026.
- ICCV 2025: [Learning to See: Advancing Spatial Understanding for Embodied Intelligence](#), 19.10.2025.
- CVPR 2025: [Embodied Intelligence for Autonomous Systems on the Horizon](#), 11.06.2025.
- CoRL 2024: [Safe and Robust Robot Learning for Operation in the Real World](#), 09.11.2024.
- ECCV 2024: [Autonomous Vehicles meet Multimodal Foundation Models](#), 29.09.2024.
- CVPR 2024: [Foundation Models for Autonomous Systems](#), 17.06.2024.
- CVPR 2023: [End-to-End Autonomous Driving: Emerging Tasks and Challenges](#), 18.06.2023.
- ICLR 2023: [Scene Representations for Autonomous Driving](#), 05.05.2023.

Recorded Talks

- [Democratizing Autonomous Driving with Latent World Models](#). ICRA Workshop on Long-term Human Motion Prediction, Vienna, 05.06.2026.
- [Democratizing Autonomous Driving with Open Science](#). ICRA Workshop on Generalization in Autonomous Driving, Vienna, 01.06.2026.
- [Specializing General-Purpose Video Diffusion Models](#). ECCV Tutorial: Recent Advances in Video Content Understanding and Generation, Milan, 30.09.2024.
- [Synthesizing Simulation Environments with Generative Models](#). CVPR Workshop on Data-Driven Autonomous Driving Simulation, Seattle, 18.06.2024.
- [Benchmarking Foundation Models for Autonomous Driving](#). CVPR Tutorial: Towards Building AGI in Autonomy and Robotics, Seattle, 18.06.2024.
- [Non-Reactive Autonomous Vehicle Simulation and Benchmarking](#). CVPR Workshop on Autonomous Driving, Seattle, 17.06.2024.
- [Reading, Writing, and Reviewing for Robotics and Computer Vision Research](#). Sogang University Applied Data Engineering Seminar, Virtual, 07.06.2023.
- [End-to-End Driving with Attention](#). ICRA Workshop on Scalable Autonomous Driving, London, 02.06.2023.
- [Imitation via Abstraction and Planning](#). ETH Computer Vision Lab, Zürich, 20.02.2023.
- [Imitation with Transformer-based Sensor Fusion for Autonomous Driving](#). University of Toronto AI in Robotics Seminar, Virtual, 28.03.2022.

Reviewing and Service

- *Journal Reviewer*: T-PAMI, IJCV, T-RO, RA-L, T-IP, T-ITS, T-IV
- *Conference Reviewer*: CVPR, ICCV, ECCV, WACV, CoRL, RSS, ICRA, IROS, NeurIPS, ICLR, IV
- *Publicity Chair*: [RSS Pioneers 2024](#)
- *Program Chair*: [ICLR 2023 SR4AD Workshop](#), [ECCV 2024 MLLMAV Workshop](#)

- Area Chair: [CoRL 2025 SAFE-ROL Workshop](#)
- Evaluator: [ELLIS PhD Program, 2022-2024](#), [IMPRS-IS PhD Program, 2023-2024](#)

Publications

All publications listed here have been accepted following peer review. For the latest publications (including pre-prints) and detailed citation statistics, see scholar.google.com.

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| 2026 | <ul style="list-style-type: none"> [1] L. Chen, C. Sima, K. Chitta, A. Loquercio, P. Luo, Y. Ma, and H. Li, "Intelligent robot manipulation requires self-directed learning," in <i>Workshop on Deployment of Foundation Models for Embodied AI (WDFM-EAI), Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2026. [2] S. Li, N. Ye, T. Li, K. Chitta, T. An, P. Su, B. Wang, H. Liu, C. Lv, and H. Li, "Optimization-guided diffusion for interactive scene generation," in <i>European Conference on Computer Vision (ECCV)</i>, 2026. [3] C. Lindenberg and K. Chitta, "Valdi: Value diffusion world models," in <i>World Models Workshop (WMW), Reinforcement Learning Conference (RLC)</i>, 2026. [4] L. Nguyen, M. Fauth, B. Jaeger, D. Dauner, M. Igl, A. Geiger, and K. Chitta, "Lead: Minimizing learner-expert asymmetry in end-to-end driving," in <i>Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2026. [5] Y. Pan, R. Qiao, L. Chen, K. Chitta, L. Pan, H. Mai, Q. Bu, H. Zhao, C. Zheng, P. Luo, and H. Li, "Agility meets stability: Versatile humanoid control with heterogeneous data," in <i>International Conference on Robotics and Automation (ICRA)</i>, 2026. [6] S. Tan, K. Chitta, Y. Chen, R. Tian, Y. You, Y. Wang, W. Luo, Y. Cao, P. Krahenbuhl, M. Pavone, and B. Ivanovic, "Latent chain-of-thought world modeling for end-to-end driving," in <i>Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2026. |
| 2025 | <ul style="list-style-type: none"> [7] W. Cao, M. Hallgarten, T. Li, D. Dauner, X. Gu, C. Wang, Y. Miron, M. Aiello, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, "Pseudo-simulation for autonomous driving," in <i>Conference on Robot Learning (CoRL)</i>, 2025. [8] M. Fauth, L. Nguyen, B. Jaeger, D. Dauner, M. Igl, A. Geiger, and K. Chitta, "Shred: Synthesizing rule-based environments for driving," in <i>Workshop on Autonomous Driving (WAD), Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2025. [9] B. Jaeger, D. Dauner, J. Beißwenger, S. Gerstenecker, K. Chitta, and A. Geiger, "Carl: Learning scalable planning policies with simple rewards," in <i>Conference on Robot Learning (CoRL)</i>, 2025. [10] L. Nguyen, M. Fauth, B. Jaeger, D. Dauner, M. Igl, A. Geiger, and K. Chitta, "Open x-av: Unifying end-to-end autonomous driving datasets," in <i>Workshop on Autonomous Driving (WAD), Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2025. [11] C. Sima, K. Chitta, Z. Yu, S. Lan, P. Luo, A. Geiger, H. Li, and J. M. Alvarez, "Centaur: Robust end-to-end autonomous driving with test-time training," in <i>Workshop on Test-time Scaling for Computer Vision, Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2025. |

- 2025 [12] J. Yang, **K. Chitta**, S. Gao, L. Chen, Y. Shao, X. Jia, H. Li, A. Geiger, X. Yue, and L. Chen, "Resim: Reliable world simulation for autonomous driving," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [13] B. Zhang, M. Golchoubian, I. Gilitschenski, B. Ivanovic, and **K. Chitta**, "Endpoint constrained trajectory optimization for driving foundation models," in *Workshop on Driving Simulation from Real-World Data (RealADSim)*, *International Conference on Computer Vision (ICCV)*, 2025.
- 2024 [14] L. Chen, P. Wu, **K. Chitta**, B. Jaeger, A. Geiger, and H. Li, "End-to-end autonomous driving: Challenges and frontiers," *Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, 2024.
- [15] **K. Chitta**, D. Dauner, and A. Geiger, "Sledge: Synthesizing driving environments with generative models and rule-based traffic," in *European Conference on Computer Vision (ECCV)*, 2024.
- [16] D. Dauner, M. Hallgarten, T. Li, X. Weng, Z. Huang, Z. Yang, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and **K. Chitta**, "Navsim: Data-driven non-reactive autonomous vehicle simulation and benchmarking," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [17] S. Gao, J. Yang, L. Chen, **K. Chitta**, Y. Qiu, A. Geiger, J. Zhang, and H. Li, "Vista: A generalizable driving world model with high fidelity and versatile controllability," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [18] M. Schneider, J. Cicvaric, A. Sauer, A. Geiger, and **K. Chitta**, "Generative dataset distillation: A new hope?" In *Workshop on the Dataset Distillation Challenge*, *European Conference on Computer Vision (ECCV)*, 2024.
- [19] C. Sima, K. Renz, **K. Chitta**, L. Chen, H. Zhang, C. Xie, J. Beißwenger, P. Luo, A. Geiger, and H. Li, "Drivelm: Driving with graph visual question answering," in *European Conference on Computer Vision (ECCV)*, 2024.
- [20] J. Yang, S. Gao, Y. Qiu, L. Chen, T. Li, B. Dai, **K. Chitta**, P. Wu, J. Zeng, P. Luo, J. Zhang, A. Geiger, Y. Qiao, and H. Li, "Generalized predictive model for autonomous driving," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [21] J. Zimmerlin, J. Beißwenger, B. Jaeger, A. Geiger, and **K. Chitta**, "Hidden biases of end-to-end driving datasets," in *Workshop on Foundation Models for Autonomous Systems (FM4AS)*, *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- 2023 [22] **K. Chitta**, A. Prakash, B. Jaeger, Z. Yu, K. Renz, and A. Geiger, "Transfuser: Imitation with transformer-based sensor fusion for autonomous driving," *Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, 2023.
- [23] D. Dauner, M. Hallgarten, A. Geiger, and **K. Chitta**, "Parting with misconceptions about learning-based vehicle motion planning," in *Conference on Robot Learning (CoRL)*, 2023.
- [24] B. Jaeger, **K. Chitta**, and A. Geiger, "Hidden biases of end-to-end driving models," in *International Conference on Computer Vision (ICCV)*, 2023.
- [25] T. Schreier, K. Renz, A. Geiger, and **K. Chitta**, "On offline evaluation of 3d object detection for autonomous driving," in *Workshop on Robustness and Reliability of Autonomous Vehicles in the Open-world (BRAVO)*, *International Conference on Computer Vision (ICCV)*, 2023.

2022	[26] N. Hanselmann, K. Renz, K. Chitta, A. Bhattacharyya, and A. Geiger, "King: Generating safety-critical driving scenarios for robust imitation via kinematics gradients," in <i>European Conference on Computer Vision (ECCV)</i>, 2022. [27] K. Renz, K. Chitta, O.-B. Mercea, A. S. Koepke, Z. Akata, and A. Geiger, "Plant: Explainable planning transformers via object-level representations," in <i>Conference on Robot Learning (CoRL)</i>, 2022.
2021	[28] K. Chitta, J. M. Alvarez, E. Haussmann, and C. Farabet, "Training data subset search with ensemble active learning," <i>Transactions on Intelligent Transportation Systems (T-ITS)</i>, 2021. [29] K. Chitta, A. Prakash, and A. Geiger, "Neat: Neural attention fields for end-to-end autonomous driving," in <i>International Conference on Computer Vision (ICCV)</i>, 2021. [30] A. Prakash, K. Chitta, and A. Geiger, "Multi-modal fusion transformer for end-to-end autonomous driving," in <i>Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2021. [31] A. Sauer, K. Chitta, J. Muller, and A. Geiger, "Projected gans converge faster," in <i>Advances in Neural Information Processing Systems (NeurIPS)</i>, 2021. [32] M. A. Weis, K. Chitta, Y. Sharma, W. Brendel, M. Bethge, A. Geiger, and A. S. Ecker, "Benchmarking unsupervised object representations for video sequences," <i>Journal of Machine Learning Research (JMLR)</i>, 2021.
2020	[33] A. Behl, K. Chitta, A. Prakash, E. Ohn-Bar, and A. Geiger, "Label efficient visual abstractions for autonomous driving," in <i>International Conference on Intelligent Robots and Systems (IROS)</i>, 2020. [34] K. Chitta, J. M. Alvarez, and M. Hebert, "Quadtree generating networks: Efficient hierarchical scene parsing with sparse convolutions," in <i>Winter Conference on Applications of Computer Vision (WACV)</i>, 2020. [35] E. Haussmann, M. Fenzi, K. Chitta, J. Ivanec, H. Xu, D. Roy, A. Mittel, N. Koumchatzky, C. Farabet, and J. M. Alvarez, "Scalable active learning for object detection," in <i>Intelligent Vehicles Symposium (IV)</i>, 2020. [36] E. Ohn-Bar, A. Prakash, A. Behl, K. Chitta, and A. Geiger, "Learning situational driving," in <i>Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2020. [37] A. Prakash, A. Behl, E. Ohn-Bar, K. Chitta, and A. Geiger, "Exploring data aggregation in policy learning for vision-based urban autonomous driving," in <i>Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2020.
2018	[38] K. Chitta, "Targeted kernel networks: Faster convolutions with attentive regularization," in <i>Workshop on Compact and Efficient Feature Representation and Learning in Computer Vision (CEFRL)</i>, <i>European Conference on Computer Vision (ECCV)</i>, 2018. [39] K. Chitta, J. M. Alvarez, and A. Lesnikowski, "Deep probabilistic ensembles: Approximate variational inference through kl regularization," in <i>Workshop on Bayesian Deep Learning (BDL)</i>, <i>Conference on Neural Information Processing Systems (NeurIPS)</i>, 2018.
2016	[40] K. Chitta and N. N. Sajjan, "A reduced region of interest based approach for facial expression recognition from static images," in <i>IEEE Region-10 Conference (TENCON)</i>, 2016.

References

- Prof. Andreas Geiger**. Co-Founder & Chief Executive Officer, KE:SAI. andreas@kesai.eu
- Prof. Marco Pavone**. Associate Professor, Stanford University. pavone@stanford.edu
- Dr. José M. Álvarez**. Director, Autonomous Vehicles Applied Research, NVIDIA. josea@nvidia.com
- Prof. Hongyang Li**. Assistant Professor, University of Hong Kong. hongyang@hku.hk
- Prof. Igor Gilitschenski**. Assistant Professor, University of Toronto. gilitschenski@cs.toronto.edu